FRAUD DETECTION AI — AUDIT REPORT

EU AI Act Compliance Assessment | Version 1.0.0

CONFIDENTIAL | ACADEMIC DRAFT

Executive Summary

This report presents a compliance audit of a machine learning-based fraud detection system against the EU Artificial Intelligence Act (Regulation 2024/1689). The system is classified as High-Risk under Article 6(2) and Article 6(3) — it profiles natural persons, which triggers mandatory high-risk classification regardless of the Annex III fraud detection exception. The audit covers Articles 6, 10, 12, 13, 14, and 15. Generated by pipeline v1.0.0 on 2026-05-14.

Overall Risk Verdict

Risk Level	Findings	Articles Reviewed	Deadline
HIGH	8 Identified	Arts. 6, 10-15, 27	2 August 2026

Findings at a Glance

#	Finding	Article	Severity	Status
1	High-Risk Classification Documented	Art. 6(4)	**LOW**	**PASS**
2	Accuracy Paradox — 99.83% meaningless accuracy	Art. 15	**HIGH**	**FAIL**
3	Dollar-Weighted Recall: 45.21%	Art. 15	**HIGH**	**FAIL**
4	High Value Fraud Recall: 50%	Art. 15	**HIGH**	**FAIL**
5	Threshold 0.3 outperforms default 0.5	Art. 15	**MEDIUM**	**WARNING**
6	93.3% features unexplainable	Art. 13	**HIGH**	**FAIL**
7	GDPR vs EU AI Act regulatory conflict	Art. 13 + GDPR 22	**HIGH**	**FAIL**
8	Human Oversight Not Documented	Art. 14	**HIGH**	**FAIL**

Section 1 — High-Risk Classification (Article 6)

Article 6(4) requires providers to document the high-risk classification assessment before placing the system on the market. This section fulfils that obligation.

Field	Detail
Annex III Reference	Annex III, Point 5(b)
Classification	HIGH-RISK
Legal Basis	Article 6(2) + Article 6(3) EU AI Act 2024/1689
Profiling Confirmed	True
Fraud Exception Note	This system IS the fraud detection exception referenced in Annex III Point 5(b). Fraud detection AI is explicitly exclud...
Supervisory Authority	BaFin + Bundesbank
Enforcement Deadline	2 August 2026
Documentation Date	2026-05-14T13:50:23

Section 2 — Model Performance (Article 15)

Article 15(1) requires an appropriate level of accuracy for the intended purpose. For fraud detection, count-based accuracy is misleading. Dollar-weighted recall is the primary compliance metric.

The Accuracy Paradox

A model predicting 'legitimate' for ALL 284,807 transactions achieves 99.83% accuracy while catching ZERO fraud. This pipeline's model achieves 99.95% accuracy — almost identical to the naive baseline. Accuracy is therefore meaningless as a compliance metric for this system.

Fraud-Specific Metrics

Metric	Value	Interpretation	Status
Accuracy	0.9995	MEANINGLESS — naive baseline is 0.9983	**WARNING**
Fraud Precision	0.9804	Of flagged transactions, 98% are real fraud	**PASS**
Fraud Recall	0.6667	Of actual fraud, 67% caught by count	**WARNING**
Fraud F1	0.7937	Balance of precision and recall	**ACCEPTABLE**
Average Precision	0.8136	Gold standard metric — 625x better than random	**PASS**
Dollar-Weighted Recall	45.21%	Only 45.21% of fraud VALUE caught	**FAIL**
Missed Fraud Value	EUR 4,234.96	Financial exposure in test set alone	**FAIL**
High Value Recall (>1000)	0.5000	50% of high-value fraud missed	**FAIL**

Confusion Matrix

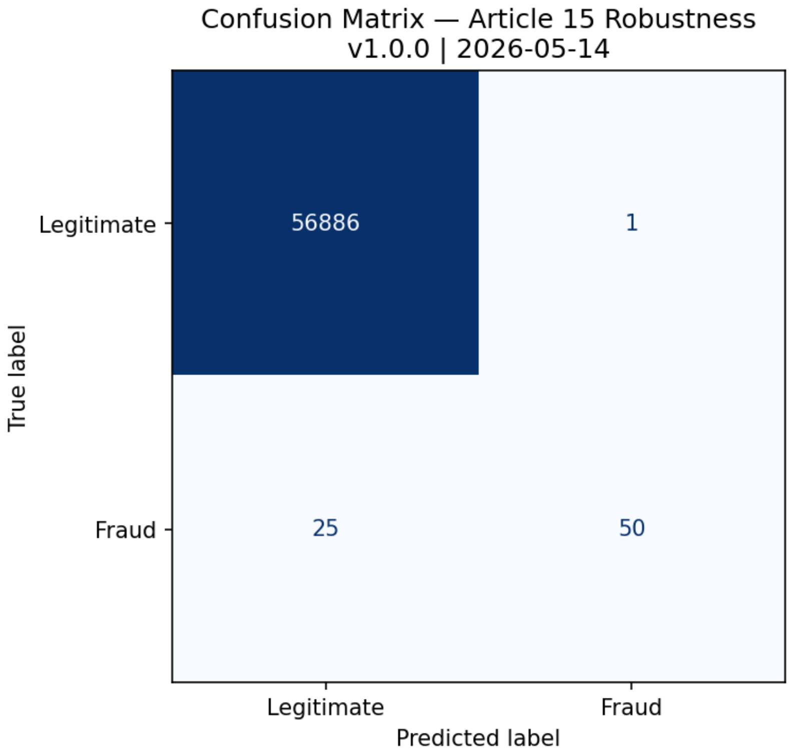

Figure 1: Confusion Matrix — Article 15 Robustness Check

Precision-Recall Curve

The Precision-Recall curve is the primary evaluation metric for imbalanced fraud detection. Average Precision of 0.8136 demonstrates the model is significantly better than random (baseline: 0.0013).

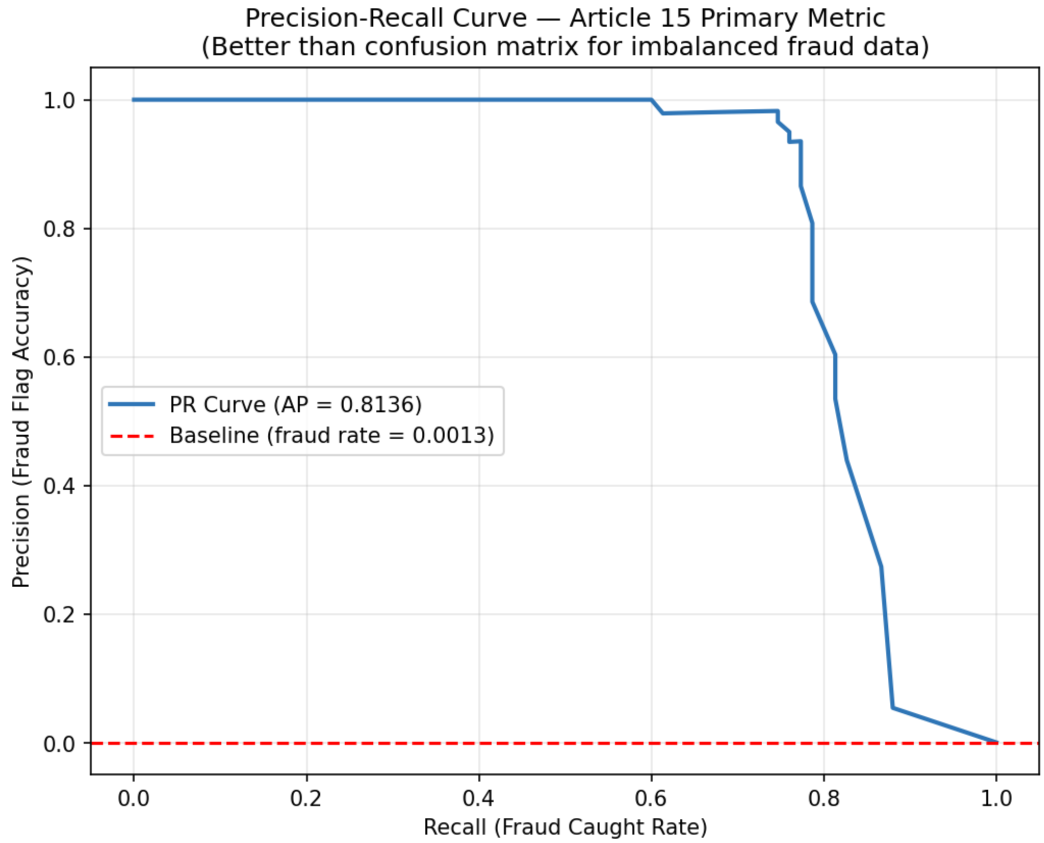

Figure 2: Precision-Recall Curve — Article 15 Primary Metric

Section 3 — Threshold Analysis (Article 15)

Article 15 requires that accuracy be appropriate for the intended purpose. The default threshold of 0.5 is not optimal for fraud detection. The table below shows performance across five thresholds. Threshold 0.3 achieves the best F1 score with negligible false positive impact.

Threshold	Precision	Recall	F1	FP Rate	Note
0.1	0.8082	0.7867	0.7973	0.0002	
0.2	0.9500	0.7600	0.8444	0.0001	
0.3	0.9825	0.7467	0.8485	0.0000	← RECOMMENDED
0.4	0.9815	0.7067	0.8217	0.0000	
0.5	0.9804	0.6667	0.7937	0.0000	

REMEDIATION: Change default threshold from 0.5 to 0.3. This increases F1 from 0.7937 to 0.8485 with zero meaningful increase in false positives. Document threshold choice formally per Article 15 requirements.

Section 4 — Transparency & GDPR Conflict (Article 13)

Article 13(1) requires that high-risk AI systems be sufficiently transparent for deployers to interpret outputs. This system has a critical transparency gap: 93.3% of features are PCA-anonymised and cannot be explained in human-readable terms.

Field	Detail
Total Features	30
PCA Anonymised (V1-V28)	28
Human Readable	2
Explainability Gap	93.3%
GDPR Conflict	True
Conflict Note	PCA anonymisation complies with GDPR data minimisation but prevents human-readable explanations required by GDPR Article 22 and EU AI Act Article 13(1...

GDPR vs EU AI Act Regulatory Conflict

This audit identifies a structural regulatory conflict with direct legal implications:

GDPR Article 5(1)(c) — Data minimisation: anonymise personal data. The bank complied by applying PCA to transaction features V1-V28.

GDPR Article 22 + EU AI Act Article 13(1) — Right to explanation: individuals subject to automated decisions must receive a meaningful explanation. PCA anonymisation makes this impossible — 'V12 had high value' means nothing to a customer.

REMEDIATION: The bank must implement a parallel explanation system using original (pre-PCA) features in a secure environment, or adopt privacy-preserving explanation techniques such as local differential privacy for SHAP values.

SHAP Feature Importance

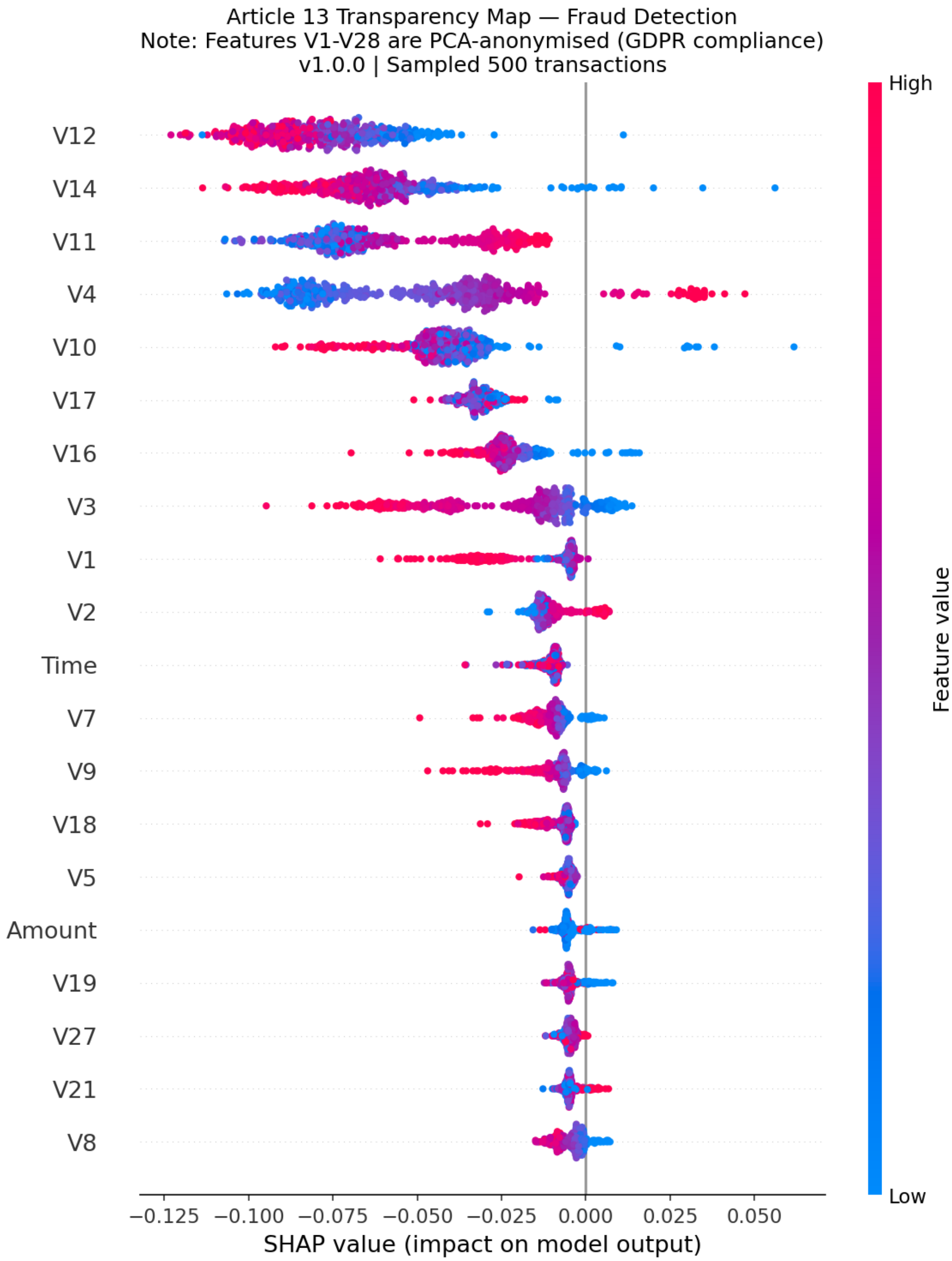

Figure 3: SHAP Summary — Article 13 Transparency Map (Note: V1-V28 are anonymised)

Section 5 — Data Governance (Article 10)

Article 10(3) requires that training and testing data be relevant, representative, and appropriate for the intended purpose.

Requirement	Status	Detail
No missing values	PASS	Zero missing values across all 284,807 records
Time-based split	PASS	Train: first 80% by time. Test: last 20%. Prevents future data leakage.
Class imbalance documented	PASS	492 fraud / 284,315 legitimate. Ratio 1:578. class_weight=balanced applied.
Imbalance handling	PASS	class_weight=balanced in Random Forest
High value subgroup tested	PASS	Transactions >EUR 1,000 audited separately
Feature anonymisation noted	WARNING	V1-V28 PCA anonymised. Creates Article 13 gap documented in Section 4.

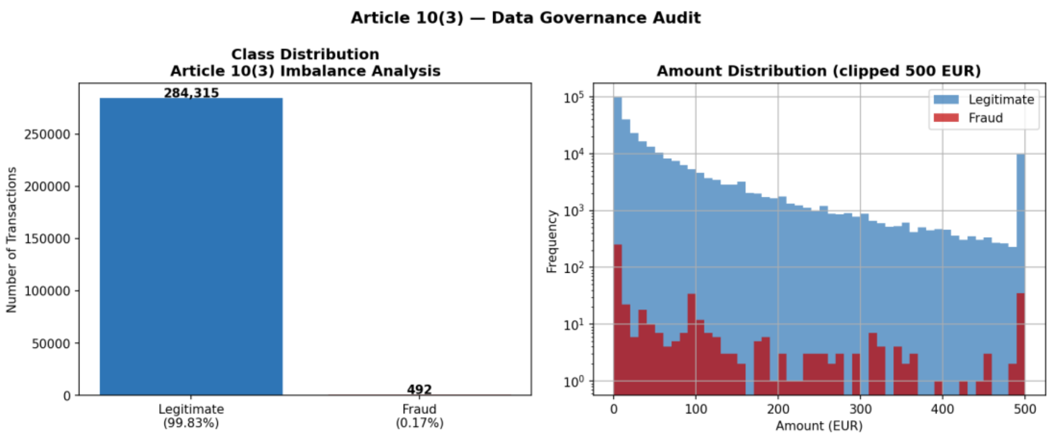

Figure 4: Data Audit — Article 10(3) Imbalance Analysis

Section 6 — Human Oversight (Article 14)

Article 14(1) requires that high-risk AI systems allow effective human oversight. No oversight mechanisms were documented for this system.

Requirement	Status	Recommendation
Human review for borderline cases	FAIL	Define review process for fraud probability 0.3-0.7
Override mechanism	FAIL	Allow fraud analysts to reverse automated blocks
Real-time monitoring	FAIL	Add model drift detection on weekly transaction batches
Shutdown procedure	FAIL	Document formal shutdown process with named responsible role

Section 7 — Remediation Roadmap

#	Action	Article	Priority	Target
1	Change decision threshold from 0.5 to 0.3	Art. 15	CRITICAL	Immediate

2	Implement dollar-weighted monitoring — EUR 4,235 missed	Art. 15	**CRITICAL**	Pre-deployment
3	Resolve GDPR/AI Act explanation conflict	Art. 13	**CRITICAL**	Pre-deployment
4	Document human oversight and override procedure	Art. 14	**HIGH**	Pre-deployment
5	Improve high-value fraud recall (currently 50%)	Art. 15	**HIGH**	30 days
6	Establish continuous risk management process	Art. 9	**HIGH**	30 days
7	Conduct Fundamental Rights Impact Assessment	Art. 27	**MEDIUM**	60 days

Section 8 — Audit Trail (Article 12)

Field	Detail
Timestamp	2026-05-14 13:51:01
Pipeline Version	1.0.0
Log File	outputs/logs/audit_log_v1.0.0.json
Seed	42 (fixed for reproducibility)
Split Method	Time-based (not random)
Classifier	RandomForestClassifier (class_weight=balanced)
Supervisory Auth	BaFin + Bundesbank (Germany)

Glossary

Term	Definition
Dollar-Weighted Recall	Percentage of total fraud VALUE in EUR detected. More meaningful than count-based recall for financial risk.
Average Precision Score	Area under Precision-Recall curve. Gold standard for imbalanced classification.
False Positive Rate	Legitimate transactions wrongly blocked. Damages customer relationships.
PCA	Principal Component Analysis. Transforms original features into anonymous components.
GDPR Article 22	Right not to be subject to automated decisions. Requires human explanation on request.
DIR	Disparate Impact Ratio. Below 0.80 indicates prima facie discrimination.
BaFin	Bundesanstalt für Finanzdienstleistungsaufsicht. German financial supervisory authority for AI Act enforcement.